

ASSIGNMENT DAY-5

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.



The main components inside a desktop pc are:

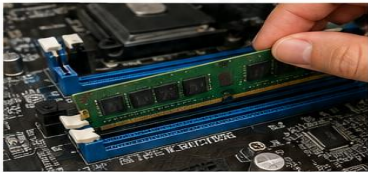
- **MOTHERBOARD:** The main circuit board of the PC which connects CPU, memory, storage and other internal components.
- **Central Processing Unit (CPU):** The brain of the computer, it executes instructions, performs calculation, and manages data flow for all the programs.
- **Memory (RAM):** Volatile memory module that temporarily store data currently in use by CPU, clears out when the PC is turn off.
- **Storage Devices (HDDs and SSDs):** The non-volatile memory module that permanently stores data even when the PC is turned off.
- **Case Fan:** Extra fan attached to the inner walls of the computer case to pull in cool air and push out hot air which keeps the whole system cool.
- **Graphic Card:** A card that draws all picture, video, and 3D graphics on your screen important for playing games.
- **Power Supply (PSU):** The metal box with a fan that plugs into the wall. It takes electricity from your house and safely powers every part inside the computer.
- **CPU Fan:** A fan sitting on top of the processor to blow away heat and keep it cool.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



DESKTOP PC DISASSEMBLY GUIDE

Step-by-step process to safely disassemble a desktop computer.

1. Unplug Power  Turn off the PC. Switch off the power supply (if available) and unplug the power cable from the back.	2. Remove Side Panel  Remove the screws at the back of the case. Slide the side panel backward and lift it off to open the case.	3. Disconnect Power Cables  Disconnect the 24-pin motherboard power cable and the 4/8-pin CPU power cable from the motherboard.
4. Disconnect Data Cables  Disconnect SATA data cables from HDD/SSD and any other connected drives or devices.	5. Remove RAM Modules  Press the clips on both sides of the RAM slot. The module will pop up. Gently remove the RAM stick.	6. Remove HDD / SSD  Remove the screws or release the tray that holds the drive. Slide out the HDD/SSD and disconnect any remaining cables.
7. Remove Expansion Cards (If any)  Remove the screw holding the expansion card. Gently pull the card out from the PCIe slot.	8. Disconnect Front Panel Cables  Carefully disconnect front panel cables (Power SW, Reset SW, HDD LED, Power LED, etc.) from the motherboard.	9. Remove CPU Cooler (If needed)  Unplug the CPU fan cable. Remove the cooler by unscrewing or releasing the clips. Clean the area if necessary.
10. Remove Motherboard  Remove all screws securing the motherboard to the case. Carefully lift the motherboard out of the case.	11. Disassembly Complete  All major components have been removed successfully. Handle each part with care and store safely.	Important Tips ✓ Always unplug power before starting. ✓ Use an anti-static wrist strap to avoid damaging components. ✓ Keep screws and small parts in a container. ✓ Handle components by the edges. ✓ Take your time and be gentle.



Note: This guide is for educational purposes. Disassemble at your own risk. Improper handling may damage components.

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.



Steps to reassemble the PC are:

1. Put the Motherboard:

placed the motherboard inside the case and fixed it with screws.

Why: This is the main part. Everything else connects to it.

2. Install the CPU:

placed the CPU in its slot carefully and locked it.

Why: The CPU is the brain, so it must be installed early.

3. Attach the CPU Fan:

fixed the fan on top of the CPU and connected its wire.

Why: It keeps the CPU cool. Without it, the PC can overheat.

4. Install RAM:

inserted the RAM sticks into the slots until they clicked.

Why: RAM is needed for the PC to start.

5. Install Hard Disk / SSD:

placed the storage drive in its place and screwed it.

Why: It stores all data and the operating system.

6. Install Graphics Card:

inserted the GPU into the slot and fixed it with screws.

Why: It should be added before connecting wires.

7. Connect Power Cables:

connected power cables to motherboard, CPU, and GPU.

Why: Power is needed to run all parts, so connect after installing them.

8. Connect Data Cables:

connected SATA cables from storage to motherboard.

Why: These cables help parts communicate.

9. Connect Front Panel Wires:

connected power button, reset button, USB wires.

Why: These are small wires, easier to connect before closing.

10. Check Everything:

checked all connections and arranged wires properly.

Why: Loose wires can cause problems.

11. Close Case and Start PC:

closed the case, plugged in power, and turned on the PC.

Why: Final step to test if everything works.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.



Safety Checklist and Tools needed before starting the PC are:

➤ **TOOLS NEEDED:**

- **Screwdriver (Phillips head)**
Why: Used to open the cabinet and remove or fix components safely.
- **Anti-static wrist strap**
Why: Prevents static electricity from damaging sensitive parts like RAM and motherboard.
- **Clean, dry cloth**
Why: Helps clean dust without harming components.
- **Small container or box**
Why: Keeps screws safe so they don't get lost.
- **Flashlight (optional)**
Why: Helps you see small parts and connectors clearly.

➤ **SAFETY PRECAUTIONS:**

- **Turn off the PC completely**
Why: Prevents electric shock and damage to components.
- **Unplug power cable**
Why: Ensures no electricity is flowing inside the system.
- **Remove all external devices**
Why: Avoids damage to ports and makes handling easier.
- **Work on a clean, flat surface**
Why: Prevents parts from falling or getting damaged.
- **Avoid touching components directly**
Why: Oils and static from hands can damage parts.
- **Handle parts by edges only**
Why: Protects delicate circuits on components.
- **Do not force any component**
Why: Prevents breaking slots or connectors.
- **Keep liquids away**
Why: Water can cause short circuits.
- **Take your time and stay organized**
Why: Reduces mistakes and makes reassembly easier.

